🎮 Player Loyalty Scoring & Reward Optimization

Analytical Report

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1. 📌 Executive Summary

This report presents a data-driven framework to evaluate player engagement and financial activity on a gaming platform and translate it into a robust loyalty scoring and reward system.

Using a simulated dataset (in absence of provided data), the analysis computes loyalty points, ranks players, and proposes an optimized bonus allocation strategy. The findings highlight imbalances in the current scoring logic and recommend improvements to enhance fairness, engagement, and long-term profitability.

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2. 🎯 Problem Statement

The platform rewards players with loyalty points based on:

- Deposits
- Withdrawals
- Gameplay activity

Top 50 players receive bonus payouts based on their performance.

Key Objectives:

1. Compute player-wise loyalty points across defined time slots
2. Rank players based on monthly performance
3. Analyze player behavior and engagement patterns
4. Design an optimal bonus distribution strategy
5. Evaluate and improve the existing loyalty formula

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3. 🧠 Data & Methodology

3.1 Data Assumptions

As no dataset was provided, a synthetic dataset was generated to simulate realistic player behavior:

- Financial activity follows skewed distributions (few high spenders, many low spenders)
- Gameplay follows count-based distributions typical of user engagement
- Time-based activity spans the full month of October

3.2 Data Preparation

- Extracted date and hour from timestamps
- Created time slots:
  - **S1:** 00:00 – 12:00
  - **S2:** 12:00 – 24:00

3.3 Feature Engineering

For each player, date, and slot:

- Total deposit and withdrawal amounts
- Number of deposit and withdrawal events
- Total games played

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4. 🧮 Loyalty Point Framework

Loyalty Score is calculated as:

Loyalty Score =
(0.01 × Deposit Amount) +
(0.005 × Withdrawal Amount) +
(0.001 × max(Deposit Count − Withdrawal Count, 0)) +
(0.2 × Games Played)

Interpretation:

- Financial contribution drives primary value

- Gameplay ensures engagement is rewarded
- Frequency adjustment encourages consistent deposits

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5. 📊 Analysis & Key Findings

5.1 Slot-Based Analysis

Player performance was evaluated across selected time slots to capture intra-day behavioral variations.

Insight:
Engagement and transaction activity vary across slots, indicating opportunities for time-based promotions and targeted incentives.

5.2 Leaderboard Dynamics

Players were ranked based on:

1. Total loyalty score
2. Total games played (tie-breaker)

Key Observations:

- Loyalty distribution is highly skewed
- A small subset of players dominates the leaderboard
- High deposit activity strongly influences ranking

5.3 Behavioral Patterns

- Majority of users exhibit low-to-moderate financial activity
- Gameplay is more evenly distributed across users
- Strong relationship exists between deposits and loyalty score

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5.4 Engagement Insights

- Consistent players tend to outperform sporadic high spenders over time

- Time-slot variation suggests peak engagement windows
- Withdrawal contribution dilutes the intended reward structure

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6. 💰 Bonus Allocation Strategy

Proposed Approach: Hybrid Weighted Model

Bonus pool: ₹50,000 allocated among top 50 players

Allocation Criteria:

- 60% weight → Loyalty Score
- 40% weight → Games Played

Justification:

This approach:

- Reduces over-reliance on financial contribution
- Rewards consistent gameplay behavior
- Promotes fairer competition among players

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7. 🔍 Advanced Analytical Insights

7.1 Player Segmentation

Clustering reveals three distinct groups:

- Low engagement users
- Moderate activity users
- High-value / highly engaged users

👉 Enables targeted retention and reward strategies

7.2 Efficiency Analysis

“Points per Game” metric highlights:

- Players who generate higher value per interaction
- Opportunity to reward efficiency over volume

7.3 Outlier Detection

A small number of players contribute disproportionately high scores, indicating:

- Potential high-value customers
- Possible scoring imbalance

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8. ⚖️ Evaluation of Current Loyalty Model

Key Limitations:

1. **Deposit Bias**
   Heavily favors high spenders over regular players
2. **Withdrawal Incentivization**
   Rewards behavior that may reduce platform revenue
3. **Lack of Skill Measurement**
   Does not account for player performance or outcomes
4. **Susceptibility to Skew**
   Extreme users dominate rankings

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9. 🚀 Recommendations

9.1 Redefine Financial Contribution

- Use **Net Contribution (Deposit – Withdrawal)**
- Align rewards with actual platform value

9.2 Improve Fairness

- Introduce normalization or caps
- Reduce dominance of extreme users

9.3 Enhance Engagement Metrics

- Include consistency indicators
- Incorporate skill-based metrics (e.g., win rate)

9.4 Personalization Strategy

- Use segmentation to create tiered rewards
- Design targeted campaigns for different user groups

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10. 📌 Key Takeaways

- Loyalty scoring is currently skewed toward financial inputs
- Engagement plays a critical but secondary role
- A hybrid reward system improves fairness and retention
- Data-driven refinement is essential for long-term success

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11. ✅ Final Conclusion

This analysis establishes a scalable and practical framework for player evaluation and reward distribution.

A balanced loyalty system must:

- Encourage engagement
- Ensure fairness
- Align with business profitability

By refining the scoring model and adopting a hybrid reward strategy, the platform can significantly improve both user retention and revenue outcomes.